Lecture 21
Refinement Types

With slides from Nadia Polikarpova

Non-trivial properties with refinement types

```
replicate :: n: Int → x: β → v: List
β
replicate :: n: Nat → x: β → { v: List β | len v =
n }
```

Example: insert

Goal type:

x: β → xs: IncList β → { v: IncList β | elems v = elems xs + [x] }

Components:

data IncList α **where**

Nil :: : IncList | }

Cons :: x: α → xs: IncList { v: α | v ≥ x }

→ : IncList [x]}

Insert solution

Goal type:

x: β → xs: IncList β → { v: IncList β | elems v = elems xs + [x] }

Solution:

```
insert x xs = match xs with
  Nil → Cons x Nil
  Cons y ys → if x ≤ y
    then Cons x (Cons y ys)
    else Cons y (insert x ys):: IncList { v: Int | v ≥ y }
```

Static Semantics

Typing rules

- Typing rules tell us how to derive typing judgments

$$\frac{premises}{Judgment}$$

Ex. Language of Expressions

$$\frac{x{:}T \in \Gamma}{\Gamma \vdash x : T} \qquad \frac{}{\Gamma \vdash N : int} \qquad \frac{\Gamma \vdash e1 : int \qquad \Gamma \vdash e2 : int}{\Gamma \vdash e1 + e2 : int}$$

Ex. Language of Expressions

$$\frac{x{:}T \in \Gamma}{\Gamma \vdash x : T} \qquad \frac{}{\Gamma \vdash N : int} \qquad \frac{\Gamma \vdash e1 : int \qquad \Gamma \vdash e2 : int}{\Gamma \vdash e1 + e2 : int}$$

Show that the following Judgment is valid

$$x{:}int, y{:}int \vdash x + (y + 5) : int$$

$$\frac{x{:}int, y{:}int \vdash x{:}int \quad x{:}int, y{:}int \vdash (y + 5) : int}{x{:}int, y{:}int \vdash x + (y + 5) : int}$$

$$\frac{\dfrac{x{:}int \in x{:}int, y{:}int}{x{:}int, y{:}int \vdash x{:}int} \qquad \dfrac{x{:}int, y{:}int \vdash y{:}int \quad x{:}int, y{:}int \vdash 5 : int}{x{:}int, y{:}int \vdash (y + 5) : int}}{x{:}int, y{:}int \vdash x + (y + 5) : int}$$

Simply Typed λ Calculus (F_1)

Basic Typing Rules

$$\frac{x{:}\tau \in \Gamma}{\Gamma \vdash x : \tau} \qquad \frac{\Gamma, x{:}\tau_1 \vdash e{:}\tau_2}{\Gamma \vdash (\lambda x{:}\tau_1\ e){:}\tau_1 \to \tau_2} \qquad \frac{\Gamma \vdash e_1{:}\tau' \to \tau \quad \Gamma \vdash e_2{:}\tau'}{\Gamma \vdash e_1 e_2{:}\tau}$$

Extensions

$$\frac{}{\Gamma \vdash N : int} \qquad \frac{\Gamma \vdash e1 : int \qquad \Gamma \vdash e2 : int}{\Gamma \vdash e1 + e2 : int} \qquad \frac{\Gamma \vdash e1 : int \qquad \Gamma \vdash e2 : int}{\Gamma \vdash e1 = e2 : bool}$$

$$\frac{\Gamma \vdash e{:}bool \qquad \Gamma \vdash e_t : \tau \qquad \Gamma \vdash e_f : \tau}{\Gamma \vdash if\ e\ then\ e_t\ else\ e_f : \tau}$$

Example

Is this a valid typing judgment?

$$\vdash (\lambda x{:}\, bool\ \lambda y{:}\, int\quad if\ x\ then\ y\ else\ y + 1){:}\, bool \to int \to int$$

How about this one?

$$\vdash (\lambda x{:}\, int\ \ \lambda y{:}\, bool\ x + y){:}\, int \to bool \to int$$

Example

What's the type of this function?

(λ f. λ x. *if* x = 1 *then* x *else* (f f (x-1)) * x)

$$\frac{x{:}\tau \in \Gamma}{\Gamma \vdash x : \tau} \qquad \frac{\Gamma, x{:}\tau_1 \vdash e{:}\tau_2}{\Gamma \vdash (\lambda x{:}\tau_1\ e){:}\tau_1 \to \tau_2} \qquad \frac{\Gamma \vdash e_1{:}\tau' \to \tau \quad \Gamma \vdash e_2{:}\tau'}{\Gamma \vdash e_1 e_2{:}\tau}$$

$$\frac{}{\Gamma \vdash N : int} \qquad \frac{\Gamma \vdash e1 : int \qquad \Gamma \vdash e2 : int}{\Gamma \vdash e1 + e2 : int} \qquad \frac{\Gamma \vdash e1 : int \qquad \Gamma \vdash e2 : int}{\Gamma \vdash e1 = e2 : bool}$$

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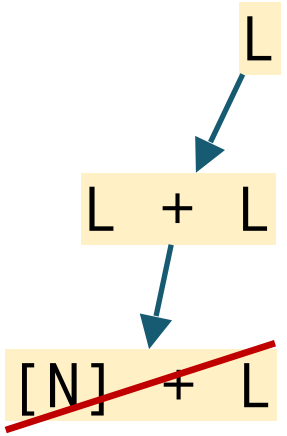
- Hint: This IS a trick question

Motivation

Goal: use deductive reasoning for top-down propagation

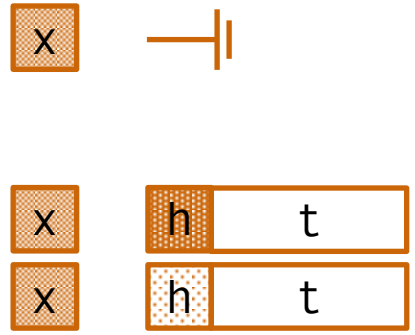
- prune unverifiable candidates early
- need synthesis-friendly verification technique!

Observation: type checkers are good at rejecting incomplete programs!

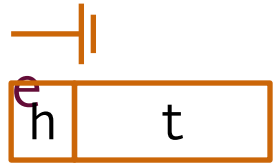


Running example

```
// Insert x into a sorted list xs
insert :: x:e  →  xs:List e  →  List e
insert x xs =
  match xs with
    Nil →              l
    Cons h t →
      if x ≤ h
        then Cons x xs
        else Cons h (insert x t)
```




```
data List e where
  Nil :: List e
  Cons :: h:e  →  t:List e  →  List e
```



Rejecting incomplete programs

[Pierce, Turner. TPLS'00]

```
// Insert x into a sorted list xs
insert :: x:e  →  xs:List e  →
List e
insert x xs =
  match xs with
    Nil → Cons xs ...
    ...
```

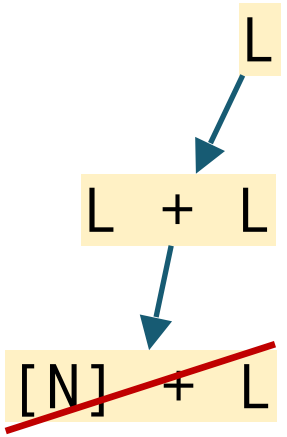
bidirectional type-checking!

Expected
e
and got
List e

Motivation

Goal: use deductive reasoning for top-down propagation

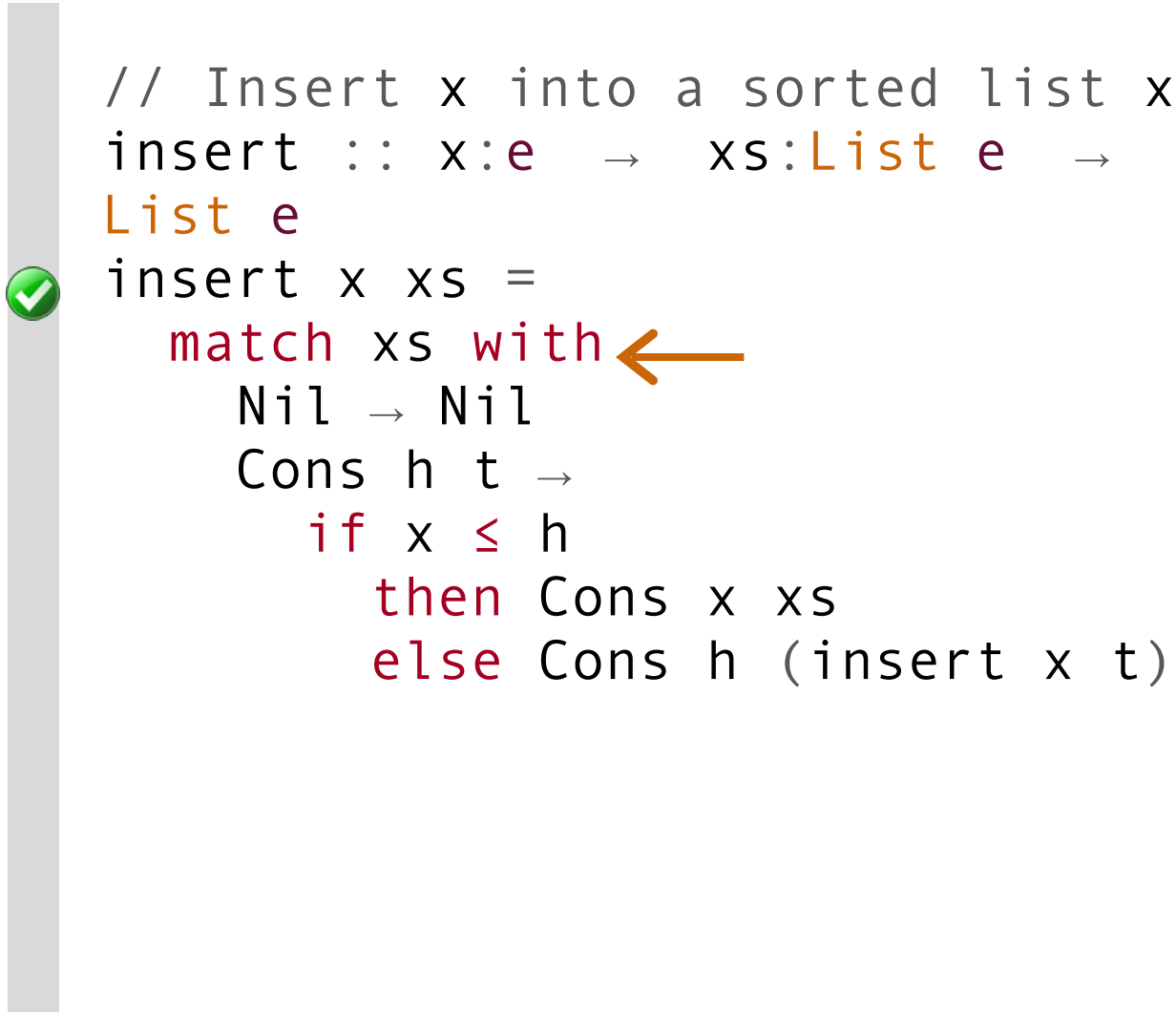
- prune unverifiable candidates early
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Observation: type checkers are good at rejecting incomplete programs!

Idea: can we use types as behavioral constraints for synthesis?

Conventional types are not enough



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List e
insert x xs =
  match xs with
    Nil → Nil
    Cons h t →
      if x ≤ h
        then Cons x xs
        else Cons h (insert x t)
```

Refinement types

[Rondon et al.'08, Kawaguchi et al.'09]

```
Nat
```

base types

```
max :: x: Int → y: Int → { v: Int | x ≤ v ∧ y ≤ v }
```

dependent function types

```
xs :: { v: List Nat }
```

polymorphic datatypes

```
data List α where
    Nil  :: { List α | len v = 0 }
    Cons :: x: α → xs: List α → { List α | len v = len xs + 1 }

List α
```

```
measure len :: List α → Int
    len Nil = 0
    len (Cons _ xs) = len xs + 1
```

Refinement types

$$e ::= \text{true} \mid \text{false} \mid n \mid e + e \mid x \mid e\ e \mid \lambda x{:}T.\ e$$

Terms

$$T ::= \{\nu{:}\,\mathrm{B} \mid e\} \mid x{:}T_1 \to T_2 \mid \alpha$$

(basic types) (function types) (type variables)

Types

$$S ::= T \mid \forall \alpha\,.\,S$$

Type schemas

T-num
$$\frac{(n = 0, 1, \ldots)}{\Gamma \vdash n :: \{\nu{:}\,\mathsf{Int} \mid \nu = n\}}$$

T-abs
$$\frac{\Gamma; x{:}T \vdash e :: T'}{\Gamma \vdash \lambda x{:}T.\ e :: T \to T'}$$

T-var
$$\frac{(x{:}\{\nu{:}B \mid P(\nu)\} \in \Gamma)}{\Gamma \vdash x :: \{\nu{:}B \mid \nu = x\}} \quad \frac{\Gamma(x) \textit{not basic}}{\Gamma \vdash x :: \Gamma(x)}$$

T-app
$$\frac{\Gamma \vdash e_1 :: x{:}T \to T' \quad \Gamma \vdash e_2 :: T}{\Gamma \vdash e_1\ e_2 :: T'[x \to e_2]}$$

Example

$$Nat \equiv \{v{:}\, Int \mid v > 0\}$$

$$\text{T-num}\ \frac{(n = 0, 1, \ldots)}{\Gamma \vdash n :: \{v{:}\,\mathrm{Int} \mid v = n\}} \qquad \text{T-abs}\ \frac{\Gamma; x{:}T \vdash e :: T'}{\Gamma \vdash \lambda x{:}T.\, e :: T \to T'}$$

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$$incr{:}\, x{:}\, Nat \to \{v{:}\, Int \mid v = x + 1\} \vdash \lambda y{:}\, Nat\,.\ incr\ y :: y{:}\, Nat \to \{v{:}\, Int \mid v = y + 1\}\,.$$

Example

$$Nat \equiv \{\nu : Int \mid \nu > 0\}$$

$$\text{T-num}\ \frac{(n = 0, 1, \ldots)}{\Gamma \vdash n :: \{\nu : \text{Int} \mid \nu = n\}} \qquad \text{T-abs}\ \frac{\Gamma; x : T \vdash e :: T'}{\Gamma \vdash \lambda x : T.\, e :: T \to T'}$$

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$$\frac{\begin{array}{c} incr : x : Nat \to \{\nu : Int \mid \nu = x + 1\}; \\ y : Nat \end{array} \vdash incr :: x : Nat \to \{\nu : Int \mid \nu = x + 1\}\ .}{\begin{array}{c} incr : x : Nat \to \{\nu : Int \mid \nu = x + 1\}; \\ y : Nat \end{array} \vdash y :: Nat}$$

$$\frac{incr : x : Nat \to \{\nu : Int \mid \nu = x + 1\}; y : Nat \vdash incr\ y :: \{\nu : Int \mid \nu = y + 1\}\ .}{incr : x : Nat \to \{\nu : Int \mid \nu = x + 1\} \vdash \lambda y : Nat\ .\ incr\ y :: y : Nat \to \{\nu : Int \mid \nu = y + 1\}\ .}$$

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$x{:}Nat \to \{v{:}Int \mid v = x + 1\};$ is not a basic type

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Basic rules by themselves are not very useful!!

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Another Example

$$Nat \equiv \{v : Int \mid v > 0\}$$

$$\text{dbl}: x:\text{Int} \to \{v:\text{Int} \mid v = 2 * x\} \vdash (\lambda y:\text{Nat}.\,\text{dbl}\ y)\ 5 :: \text{Nat}$$

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Subtyping

Intuitively, T' is a subtype of T if all values of type T' also belong to T

- written $T' <: T$
- e.g. Nat $<:$ Int or $\{\nu\text{:Int} \mid \nu = 5\} <:$ Nat

Defined via inference rules:

$$\text{Sub-base}\ \frac{[\![\Gamma]\!] \wedge e' \Rightarrow e}{\Gamma \vdash \{\nu\text{: } B \mid e'\} <: \{\nu\text{: } B \mid e\}}$$

$$\text{Sub-fun}\ \frac{\Gamma \vdash T_1 <: T_1' \qquad \Gamma; x{:}T_1 \vdash T_2' <: T_2}{\Gamma \vdash x{:}T'_1 \to T_2' <: x{:}T_1 \to T_2}$$

$$\frac{\Gamma \vdash e{:}T' \quad \Gamma \vdash T' <: T}{\Gamma \vdash e{:}T}$$

Constraints from the environment

We saw the expression $[\![\Gamma]\!]$ in the rule

- This extracts constraints from the environment

$$[\![\Gamma]\!] = \bigwedge \{ e \mid e \in \Gamma \} \wedge \bigwedge \{ [\![e[v \to x]]\!] \mid x : \{ v : B \mid e \} \in \Gamma \}$$

- The environment can store constraints e, which get included.
- The environment also contains variables $x : \{ v : B \mid e \}$.
  All their constraints must also be included, with suitable renaming inside the constraint.

Example

$$Nat \equiv \{v{:}\, Int \mid v > 0\}$$

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$$\text{T-var}\ \frac{(x{:}\{v{:}\,B \mid P(v)\} \in \Gamma)}{\Gamma \vdash x :: \{v{:}\,B \mid v = x\}} \quad \frac{\Gamma(x) \text{not basic}}{\Gamma \vdash x :: \Gamma(x)} \qquad \text{T-app}\ \frac{\Gamma \vdash e_1 :: x{:}T \to T' \quad \Gamma \vdash e_2 :: T}{\Gamma \vdash e_1\ e_2 :: T'[x \to e_2]}$$

$$\text{Sub-fun}\ \frac{\Gamma \vdash T_1 <: T_1' \qquad \Gamma; x{:}T_1 \vdash T_2' <: T_2}{\Gamma \vdash\ x{:}T'_1 \to T_2' <:\ x{:}T_1 \to T_2}$$

$$\text{Sub-base}\ \frac{[\![\Gamma]\!] \wedge e' \Rightarrow e}{\Gamma \vdash\ \{v{:}\ B \mid e'\} <:\ \{v{:}\ B \mid e\}}$$

$$\frac{\Gamma \vdash e{:}T' \quad \Gamma \vdash T' <: T}{\Gamma \vdash e{:}T}$$

$$y > 0 \wedge v = y \Rightarrow v > 0$$

$$\frac{incr{:}x{:}Nat \to \{v{:}\,Int \mid v = x+1\};\ y{:}Nat \vdash y :: \{v{:}\,Int \mid v = x\}}{incr{:}x{:}Nat \to \{v{:}\,Int \mid v = x+1\};\ y{:}Nat \vdash \{v{:}\,Int \mid v = y\} <: \{v{:}\,Int}$$

$$\frac{}{incr{:}x{:}Nat \to \{v{:}\,Int \mid v = x+1\};\ y{:}Nat \vdash y :: Na}$$

Example

$$Nat \equiv \{v{:}\, Int \mid v > 0\}$$

T-num $$\frac{(n = 0, 1, \ldots)}{\Gamma \vdash n :: \{v{:}\,\mathrm{Int} \mid v = n\}}$$

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$$incr{:}x{:}Nat \to \{v{:}Int \mid v = x + 1\};\ y{:}Nat \vdash y :: Na$$

Another Example

$$Nat \equiv \{\nu : Int \mid \nu > 0\}$$

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T-abs $$\frac{\Gamma; x : T \vdash e :: T'}{\Gamma \vdash \lambda x : T.\, e :: T \to T'}$$

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$$\frac{\Gamma \vdash e : T' \quad \Gamma \vdash T' <: T}{\Gamma \vdash e : T}$$

$$\frac{\text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\} \vdash (\lambda y : \text{Nat}.\ \text{dbl}\ y) :: Nat \to Nat \qquad \text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\} \vdash 5 :: Nat}{\text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\} \vdash (\lambda y : \text{Nat}.\ \text{dbl}\ y)\ 5 :: \text{Nat}}$$

Another Example

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$$\frac{\Gamma \vdash e : T' \qquad \Gamma \vdash T' <: T}{\Gamma \vdash e : T}$$

$$\frac{\text{dbl} : x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\} \vdash 5 :: \{\nu : Int \mid \nu = 5\} \qquad \dfrac{\nu = 5 \Rightarrow \nu > 0}{\{\nu : Int \mid \nu = 5\} <: \{\nu : Int \mid \nu > 0\}}}{\text{dbl} : x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\} \vdash (\lambda y : \text{Nat}.\ \text{dbl}\ y) :: Nat \to Nat \qquad \text{dbl} : x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\} \vdash 5 :: Nat}$$

$$\text{dbl} : x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\} \vdash (\lambda y : \text{Nat}.\ \text{dbl}\ y)\ 5 :: \text{Nat}$$

Another Example

$$Nat \equiv \{\nu : Int \mid \nu > 0\}$$

$$\text{T-num}\ \frac{(n = 0, 1, \ldots)}{\Gamma \vdash n :: \{\nu : \text{Int} \mid \nu = n\}} \qquad \text{T-abs}\ \frac{\Gamma; x : T \vdash e :: T'}{\Gamma \vdash \lambda x : T.\, e :: T \to T'}$$

$$\text{T-var}\ \frac{(x : \{\nu : B \mid P(\nu)\} \in \Gamma) \quad \Gamma(x) \text{not basic}}{\Gamma \vdash x :: \{\nu : B \mid \nu = x\}} \qquad \text{T-app}\ \frac{\Gamma \vdash e_1 :: x : T \to T' \quad \Gamma \vdash e_2 :: T}{\Gamma \vdash e_1\ e_2 :: T'[x \to e_2]}$$

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$$\frac{\Gamma \vdash e : T' \quad \Gamma \vdash T' <: T}{\Gamma \vdash e : T}$$

$$\frac{\nu = 5 \Rightarrow \nu > 0}{\{\nu : Int \mid \nu = 5\} <: \{\nu : Int \mid \nu > 0\}}$$

$$\frac{\text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\} \vdash 5 :: \{\nu : Int \mid \nu = 5\} \qquad \text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\} \vdash 5 :: Nat}{\text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\} \vdash (\lambda y : \text{Nat}.\ \text{dbl}\ y) :: Nat \to Nat}$$

$$\text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\} \vdash (\lambda y : \text{Nat}.\ \text{dbl}\ y)\ 5 :: \text{Nat}$$

Another Example

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$$\frac{\Gamma \vdash e{:}T' \quad \Gamma \vdash T' <: T}{\Gamma \vdash e{:}T}$$

$$\text{dbl}{:}\ x{:}\text{Int} \to \{\nu{:}\text{Int} \mid \nu = 2 * x\}; y{:}Nat \vdash \text{dbl} :: Nat \to Nat \qquad \text{dbl}{:}\ x{:}\text{Int} \to \{\nu{:}\text{Int} \mid \nu = 2 * x\}; y{:}Nat \vdash \text{y} :: Nat$$

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$$\boxed{\frac{\Gamma \vdash e : T' \qquad \Gamma \vdash T' <: T}{\Gamma \vdash e : T}}$$

$\Gamma \equiv$

$$\text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\}; y : Nat \vdash \text{dbl} :: Int \to \{\nu : \text{Int} \mid \nu = 2 * x\} \qquad \Gamma \vdash Int \to \{\nu : \text{Int} \mid \nu = 2 * x\} <: Nat \to Nat$$

$$\text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\}; y : Nat \vdash \text{dbl} :: Nat \to Nat \qquad \text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\}; y : Nat \vdash y :: Nat$$

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Another Example

$$Nat \equiv \{\nu : Int \mid \nu > 0\}$$

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$$\frac{\Gamma \vdash e : T' \quad \Gamma \vdash T' <: T}{\Gamma \vdash e : T}$$

$\Gamma \equiv$

$$\frac{\Gamma \vdash Nat <: Int \qquad \Gamma; x : Nat \vdash \{\nu : \text{Int} \mid \nu = 2 * x\} <: Nat}{\text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\}; y : Nat \vdash \text{dbl} :: Int \to \{\nu : \text{Int} \mid \nu = 2 * x\} \quad \Gamma \vdash x : Int \to \{\nu : \text{Int} \mid \nu = 2 * x\} <: Nat \to Nat}$$

$$\frac{\text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\}; y : Nat \vdash \text{dbl} :: Nat \to Nat \qquad \text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\}; y : Nat \vdash \text{y} :: Nat}{\text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\}; y : Nat \vdash \text{dbl}\ y :: Nat}$$

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Another Example

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$$\frac{\Gamma \vdash e : T' \quad \Gamma \vdash T' <: T}{\Gamma \vdash e : T}$$

$\Gamma \equiv$

$$\frac{\Gamma \vdash Nat <: Int \qquad \Gamma; x : \{\nu : Int \mid \nu > 0\} \vdash \{\nu : \text{Int} \mid \nu = 2 * x\} <: Nat}{\text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\}; y : Nat \vdash \text{dbl} :: Int \to \{\nu : \text{Int} \mid \nu = 2 * x\} \qquad \Gamma \vdash x : Int \to \{\nu : \text{Int} \mid \nu = 2 * x\} <: Nat \to Nat}$$

$$\frac{\text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\}; y : Nat \vdash \text{dbl} :: Nat \to Nat \qquad \text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\}; y : Nat \vdash \text{y} :: Nat}{\text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\}; y : Nat \vdash \text{dbl}\ y :: Nat}$$

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$$\text{Sub-fun}\ \frac{\Gamma \vdash T_1 <: T_1' \qquad \Gamma; x : T_1 \vdash T_2' <: T_2}{\Gamma \vdash\ x : T'_1 \to T_2' <:\ x : T_1 \to T_2}$$

$$\text{Sub-base}\ \frac{[\![\Gamma]\!] \wedge e' \Rightarrow e}{\Gamma \vdash\ \{\nu :\ B \mid e'\} <:\ \{\nu :\ B \mid e\}}$$

$$\frac{\Gamma \vdash e : T' \quad \Gamma \vdash T' <: T}{\Gamma \vdash e : T}$$

$$y > 0 \wedge x > 0 \wedge \nu = 2 * x \Rightarrow \nu > 0$$

$\Gamma \equiv$

$$\frac{\Gamma \vdash Nat <: Int \qquad \Gamma; x : \{\nu : Int \mid \nu > 0\} \vdash \{\nu : \text{Int} \mid \nu = 2 * x\} <: Nat}{\text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\}; y : Nat \vdash \text{dbl} :: Int \to \{\nu : \text{Int} \mid \nu = 2 * x\} \quad \Gamma \vdash x : Int \to \{\nu : \text{Int} \mid \nu = 2 * x\} <: Nat \to Nat}$$

$$\frac{\text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\}; y : Nat \vdash \text{dbl} :: Nat \to Nat \qquad \text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\}; y : Nat \vdash \text{y} :: Nat}{\text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\}; y : Nat \vdash \text{dbl}\ y :: Nat}$$

$$\frac{\text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\}; y : Nat \vdash \text{dbl}\ y :: Nat}{\text{dbl}: x : \text{Int} \to \{\nu : \text{Int} \mid \nu = 2 * x\} \vdash (\lambda y : \text{Nat}.\ \text{dbl}\ y) :: Nat \to Nat}$$

Branches

$$\frac{\Gamma \vdash e : bool \quad \Gamma \vdash e_t : \tau \quad \Gamma \vdash e_f : \tau}{\Gamma \vdash if\ e\ then\ e_t\ else\ e_f : \tau}$$

$$\frac{\Gamma \vdash e_1 :: Bool \quad \Gamma; e_1 \vdash e_2 :: T \quad \Gamma; \neg e_1 \vdash e_2 :: T}{\Gamma \vdash if\ e_1\ then\ e_2\ else\ e_3 :: T}$$

Example:

- $\lambda x : Int\ if\ x > 0\ then\ x\ else\ -x\ : Int \rightarrow Nat$

Branches

T-num $\dfrac{(n = 0, 1, \ldots)}{\Gamma \vdash n :: \{\nu\colon \mathrm{Int} \mid \nu = n\}}$

T-var $\dfrac{(x\colon \{\nu\colon B \mid P(\nu)\} \in \Gamma) \quad \Gamma(x) \text{not basic}}{\Gamma \vdash x :: \{\nu\colon B \mid \nu = x\}} \quad \dfrac{}{\Gamma \vdash x :: \Gamma(x)}$

$$\frac{\Gamma \vdash e_1 :: Bool \quad \Gamma; e_1 \vdash e_2 :: T \quad \Gamma; \neg e_1 \vdash e_2 :: T}{\Gamma \vdash if e_1 \; then \; e_2 \; else \; e_3 :: T}$$

T-abs $\dfrac{\Gamma; x\colon T \vdash e :: T'}{\Gamma \vdash \lambda x\colon T.\, e :: T \to T'}$

T-app $\dfrac{\Gamma \vdash e_1 :: x\colon T \to T' \quad \Gamma \vdash e_2 :: T}{\Gamma \vdash e_1 \; e_2 :: T'[x \to e_2]}$

$$(-) :: t\colon Int \to \{\nu\colon Int \mid \nu = -t\}$$

$$(>) :: Int \to Int \to Bool$$

Sub-fun $\dfrac{\Gamma \vdash T_1 <: T_1' \quad \Gamma; x\colon T_1 \vdash T_2' <: T_2}{\Gamma \vdash \; x\colon T'_1 \to T_2' <:\; x\colon T_1 \to T_2}$

Sub-base $\dfrac{[\![\Gamma]\!] \wedge e' \Rightarrow e}{\Gamma \vdash \; \{\nu\colon B \mid e'\} <: \{\nu\colon B \mid e\}}$

$$\frac{\Gamma \vdash e\colon T' \quad \Gamma \vdash T' <: T}{\Gamma \vdash e\colon T}$$

$x\colon Int \vdash x > 0$

$x\colon Int; x > 0 \vdash x\colon Nat$

$x\colon Int; x \leq 0 \vdash -x\colon Nat$

$$x\colon Int \vdash if \; x > 0 \; then \; x \; else \; -x \; :Nat$$

$$\varnothing \vdash \lambda x\colon Int \; if \; x > 0 \; then \; x \; else \; -x \; :Int \to Nat$$

Branches

T-num $\dfrac{(n = 0, 1, \ldots)}{\Gamma \vdash n :: \{\nu\colon \mathrm{Int} \mid \nu = n\}}$

T-abs $\dfrac{\Gamma; x\colon T \vdash e :: T'}{\Gamma \vdash \lambda x\colon T.\, e :: T \to T'}$

Sub-fun $\dfrac{\Gamma \vdash T_1 <\colon T_1' \qquad \Gamma; x\colon T_1 \vdash T_2' <\colon T_2}{\Gamma \vdash\ x\colon T'_1 \to T_2' <\colon\ x\colon T_1 \to T_2}$

T-var $\dfrac{(x\colon \{\nu\colon B \mid P(\nu)\} \in \Gamma)}{\Gamma \vdash x :: \{\nu\colon B \mid \nu = x\}} \qquad \dfrac{\Gamma(x) \text{not basic}}{\Gamma \vdash x :: \Gamma(x)}$

T-app $\dfrac{\Gamma \vdash e_1 :: x\colon T \to T' \qquad \Gamma \vdash e_2 :: T}{\Gamma \vdash e_1\ e_2 :: T'[x \to e_2]}$

Sub-base $\dfrac{[\![\Gamma]\!] \wedge e' \Rightarrow e}{\Gamma \vdash\ \{\nu\colon\ B \mid e'\} <\colon \{\nu\colon\ B \mid e\}}$

$$\frac{\Gamma \vdash e\colon T' \qquad \Gamma \vdash T' <\colon T}{\Gamma \vdash e\colon T}$$

$$\frac{\Gamma \vdash e_1 :: Bool \qquad \Gamma; e_1 \vdash e_2 :: T \qquad \Gamma; \neg e_1 \vdash e_2 :: T}{\Gamma \vdash if e_1\ then\ e_2\ else\ e_3 :: T}$$

$$(-) :: t\colon Int \to \{\nu\colon Int \mid \nu = -t\}$$

$$(>) :: Int \to Int \to Bool$$

$$\frac{\dfrac{x\colon Int \vdash (>)\colon Int \to Int \to Bool \qquad x\colon Int \vdash x\colon Int}{x\colon Int \vdash (>)x\colon Int \to Bool} \qquad x\colon Int \vdash 0\colon Int}{x\colon Int \vdash (>)x\ \ 0\colon Bool}$$

$$x\colon Int; x > 0 \vdash x\colon Nat$$

$$x\colon Int; x \leq 0 \vdash -x\colon Nat$$

$$x\colon Int \vdash if\ x > 0\ then\ x\ else\ -x\ \colon Nat$$

$$\varnothing \vdash \lambda x\colon Int\ if\ x > 0\ then\ x\ else\ -x\ \colon Int \to Nat$$

Branches

T-num $\dfrac{(n = 0, 1, \ldots)}{\Gamma \vdash n :: \{\nu\colon \mathrm{Int} \mid \nu = n\}}$

T-abs $\dfrac{\Gamma; x\colon T \vdash e :: T'}{\Gamma \vdash \lambda x\colon T.\, e :: T \to T'}$

Sub-fun $\dfrac{\Gamma \vdash T_1 <\colon T_1' \qquad \Gamma; x\colon T_1 \vdash T_2' <\colon T_2}{\Gamma \vdash\ x\colon T'_1 \to T_2' <\colon\ x\colon T_1 \to T_2}$

T-var $\dfrac{(x\colon \{\nu\colon B \mid P(\nu)\} \in \Gamma)}{\Gamma \vdash x :: \{\nu\colon B \mid \nu = x\}} \qquad \dfrac{\Gamma(x) \text{not basic}}{\Gamma \vdash x :: \Gamma(x)}$

T-app $\dfrac{\Gamma \vdash e_1 :: x\colon T \to T' \qquad \Gamma \vdash e_2 :: T}{\Gamma \vdash e_1\ e_2 :: T'[x \to e_2]}$

Sub-base $\dfrac{[\![\Gamma]\!] \wedge e' \Rightarrow e}{\Gamma \vdash\ \{\nu\colon\ B \mid e'\} <\colon \{\nu\colon\ B \mid e\}}$

$$\frac{\Gamma \vdash e\colon T' \quad \Gamma \vdash T' <\colon T}{\Gamma \vdash e\colon T}$$

$$\frac{\Gamma \vdash e_1 :: Bool \quad \Gamma; e_1 \vdash e_2 :: T \quad \Gamma; \neg e_1 \vdash e_2 :: T}{\Gamma \vdash if e_1\ then\ e_2\ else\ e_3 :: T}$$

$$(-) :: t\colon Int \to \{\nu\colon Int \mid \nu = -t\}$$

$$(>) :: Int \to Int \to Bool$$

$$\frac{\dfrac{x\colon Int; x \le 0 \vdash (-)\colon t\colon Int \to \{\nu\colon Int \mid \upsilon = -t\} \qquad x\colon Int; x \le 0 \vdash x\colon Int}{x\colon Int; x \le 0 \vdash -x\colon \{\nu\colon Int \mid \upsilon = -x\}} \qquad x\colon Int; x \le 0 \vdash \{\nu\colon Int \mid \upsilon = -x\} <\colon \{\nu\colon Int \mid \upsilon \ge$$

$x\colon Int \vdash x > 0 \qquad x\colon Int; x > 0 \vdash x\colon Nat \qquad x\colon Int; x \le 0 \vdash -x\colon Nat$

$$x\colon Int \vdash if\ x > 0\ then\ x\ else\ -x\ :Nat$$

$$\varnothing \vdash \lambda x\colon Int\ if\ x > 0\ then\ x\ else\ -x\ :Int \to Nat$$

Conventional types are not enough



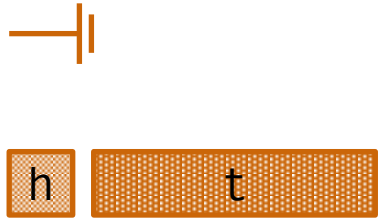
```
// Insert x into a sorted list xs
insert :: x:e  →  xs:List e  →
List e
insert x xs =
  match xs with
    Nil → Nil
    Cons h t →
      if x ≤ h
        then Cons x xs
        else Cons h (insert x t)
```

Code has a bug, but type checker still accepts it.

Refinement types

```
data SList e where
  Nil :: SList e
  Cons :: h:e →
          t:SList {v:e | v ≥ h} →
          SList e
```

sorted lists



Refinement types as specs

[Rondon et al. PLDI'08]

```
// Insert x into a sorted list xs
insert :: x:e  →  xs:SList e  →
          {v:SList e | elems v = elems xs ∪ {x}}
insert x xs =
  match xs with
    Nil → Nil
    Cons h t →
      if x ≤ h
        then Cons x xs
        else Cons h (insert x t)
```

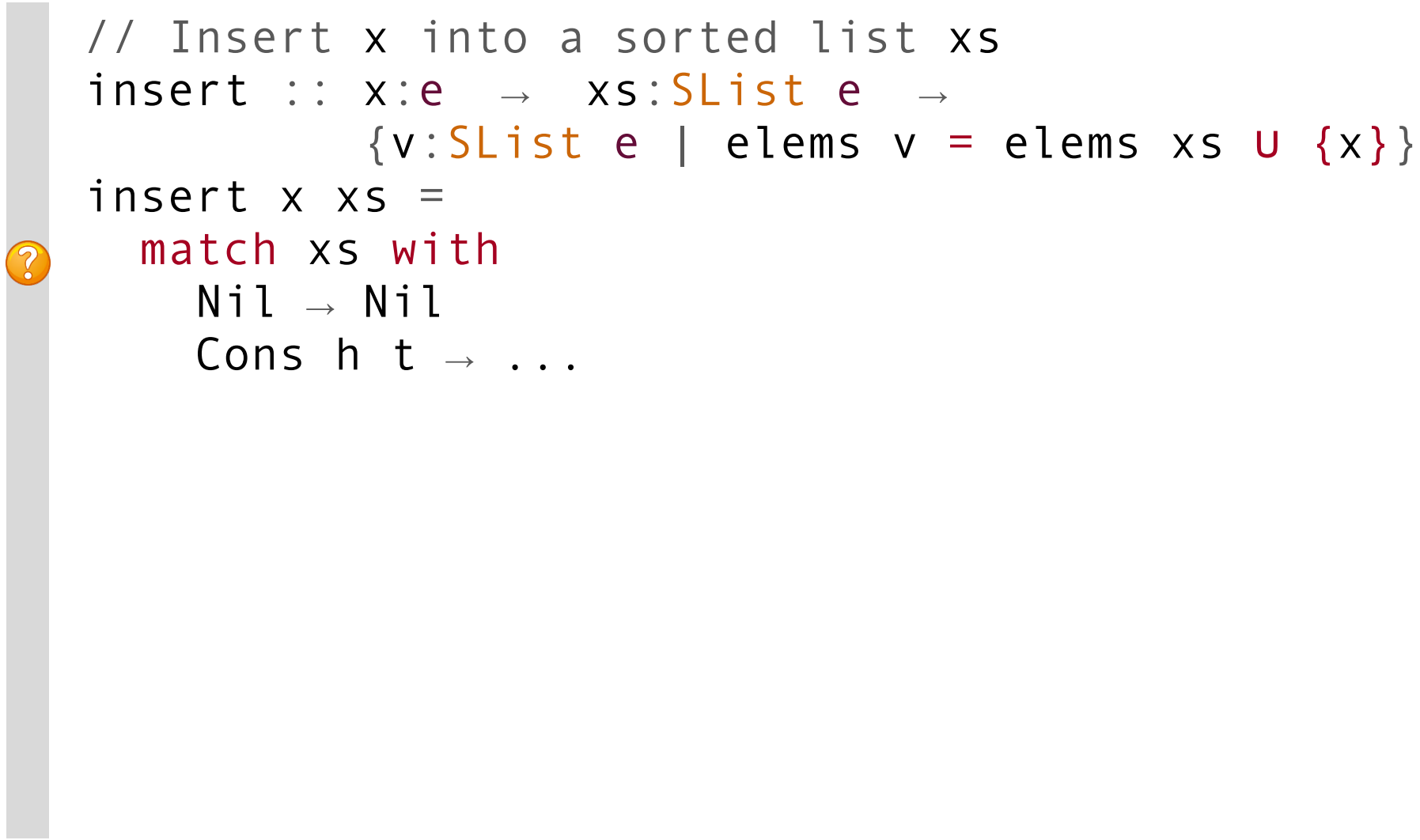
Expected

{v:SList e|elems v = elems xs ∪ {x}}

and got

{v:SList e|elems xs ⊆ elems v}

Incomplete programs?



```
// Insert x into a sorted list xs
insert :: x:e  →  xs:SList e  →
          {v:SList e | elems v = elems xs ∪ {x}}
insert x xs =
  match xs with
    Nil → Nil
    Cons h t → ...
```